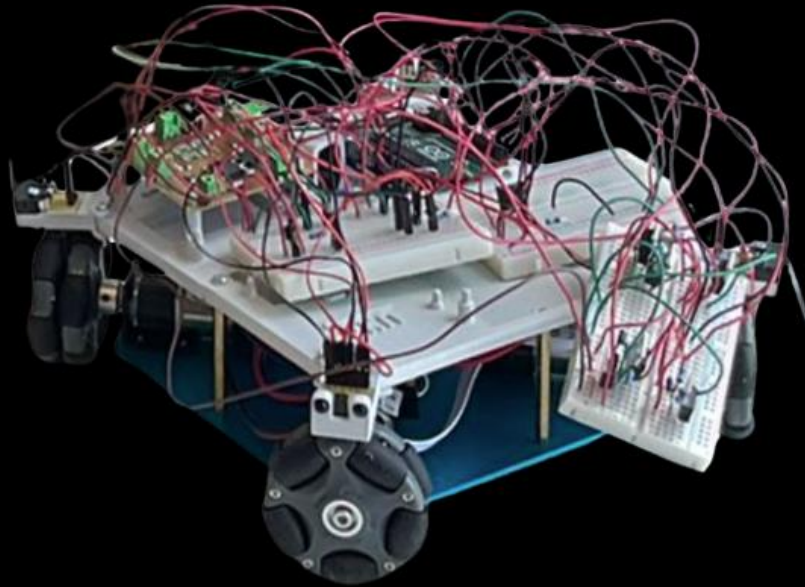
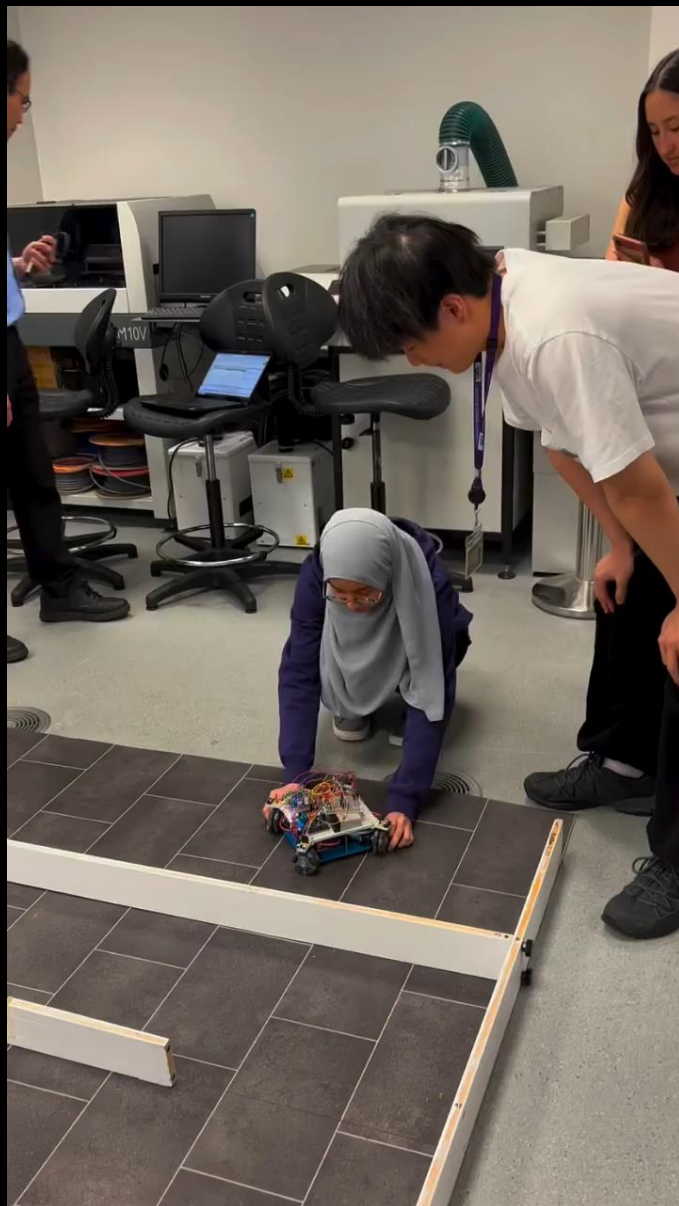


OPEN DAY ROBOTS

YOUR OPPORTUNITY TO LEARN ROBOTICS IN 90 MINS





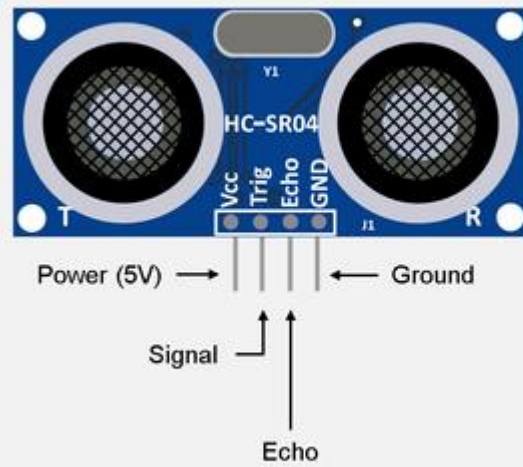




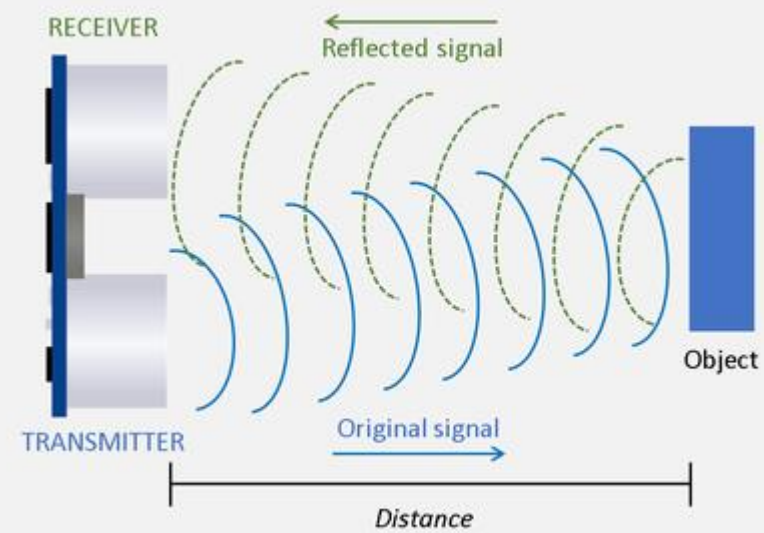








Teachwithict.com









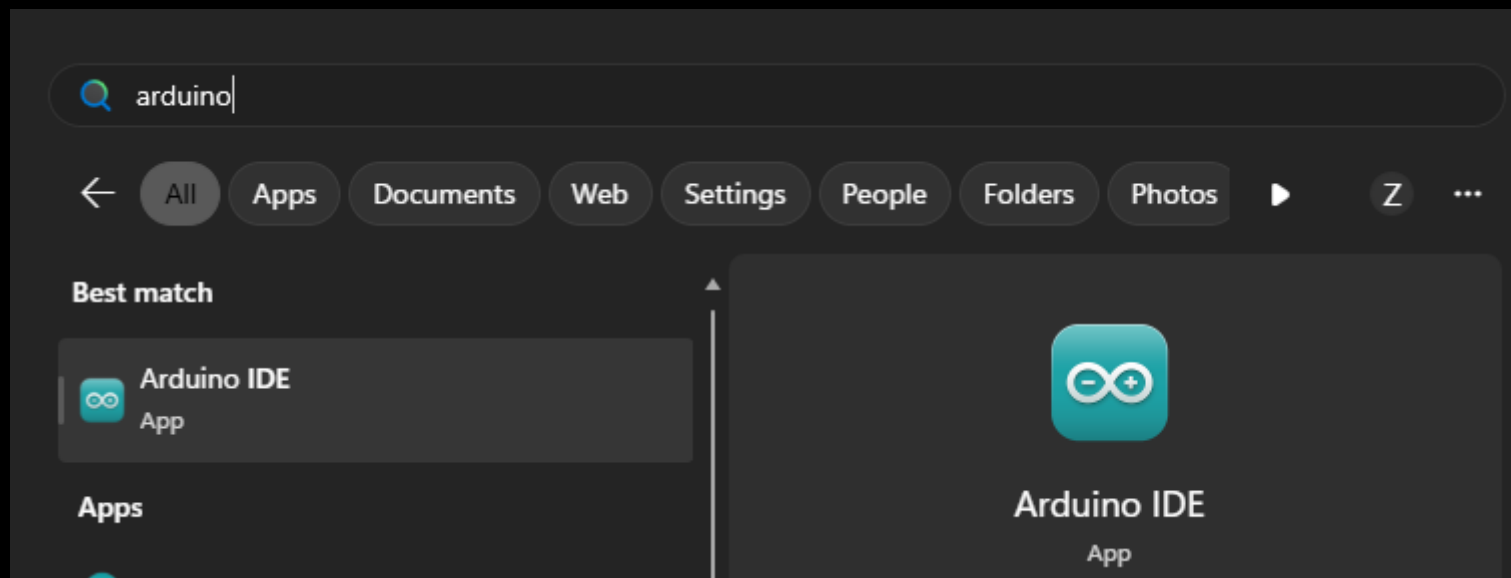


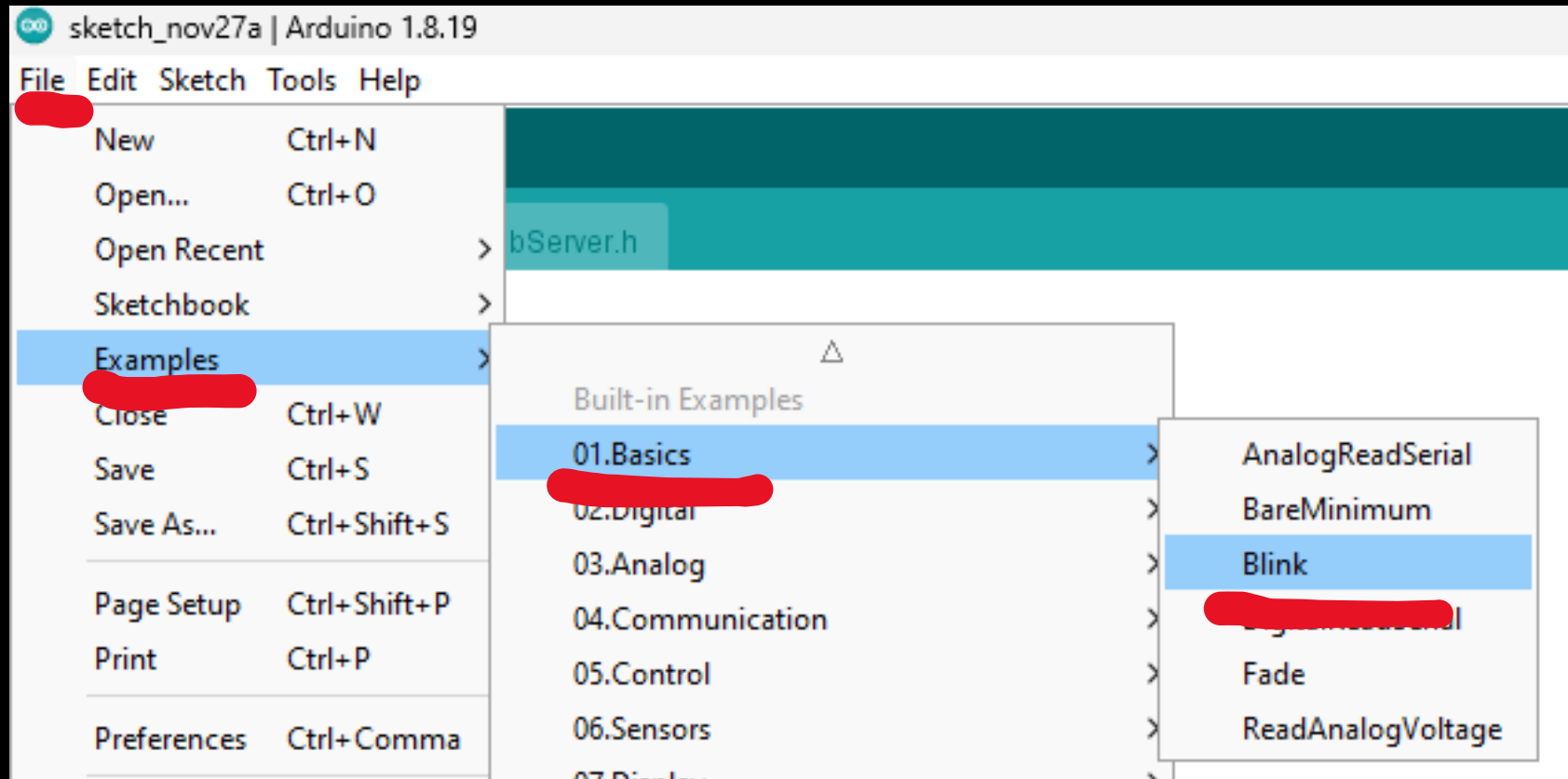
Microcontroller
Used to control electronics



BRAIN = empty







```
Blink | Arduino 1.8.19
File Edit Sketch Tools Help

✓ → 📄 ⬆ ⬇

Blink
/*
  Blink

  Turns an LED on for one second, then off for one second, repeatedly.

  Most Arduinos have an on-board LED you can control. On the UNO, MEGA and ZERO
  it is attached to digital pin 13, on MKR1000 on pin 6. LED_BUILTIN is set to
  the correct LED pin independent of which board is used.
  If you want to know what pin the on-board LED is connected to on your Arduino
  model, check the Technical Specs of your board at:
  https://www.arduino.cc/en/Main/Products

  modified 8 May 2014
  by Scott Fitzgerald
  modified 2 Sep 2016
  by Arturo Guadalupi
  modified 8 Sep 2016
  by Colby Newman

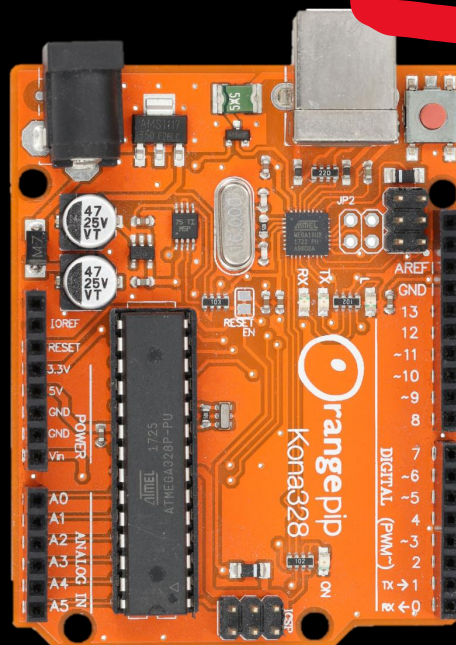
  This example code is in the public domain.

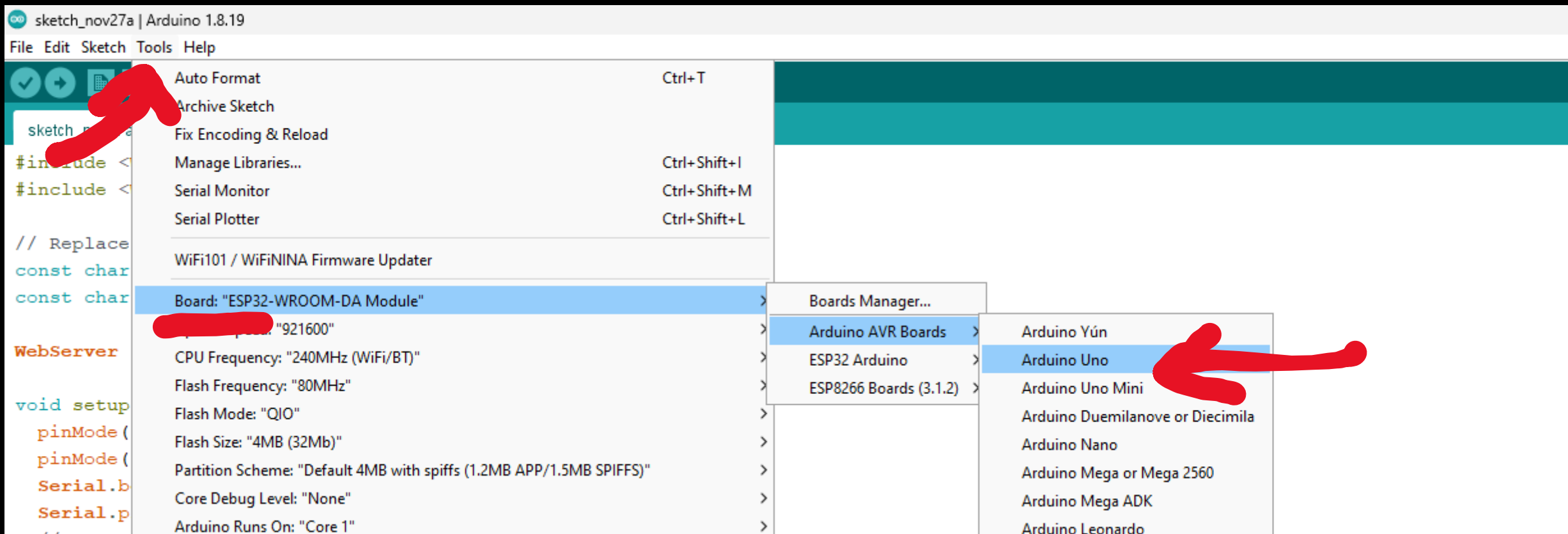
  https://www.arduino.cc/en/Tutorial/BuiltInExamples/Blink
*/

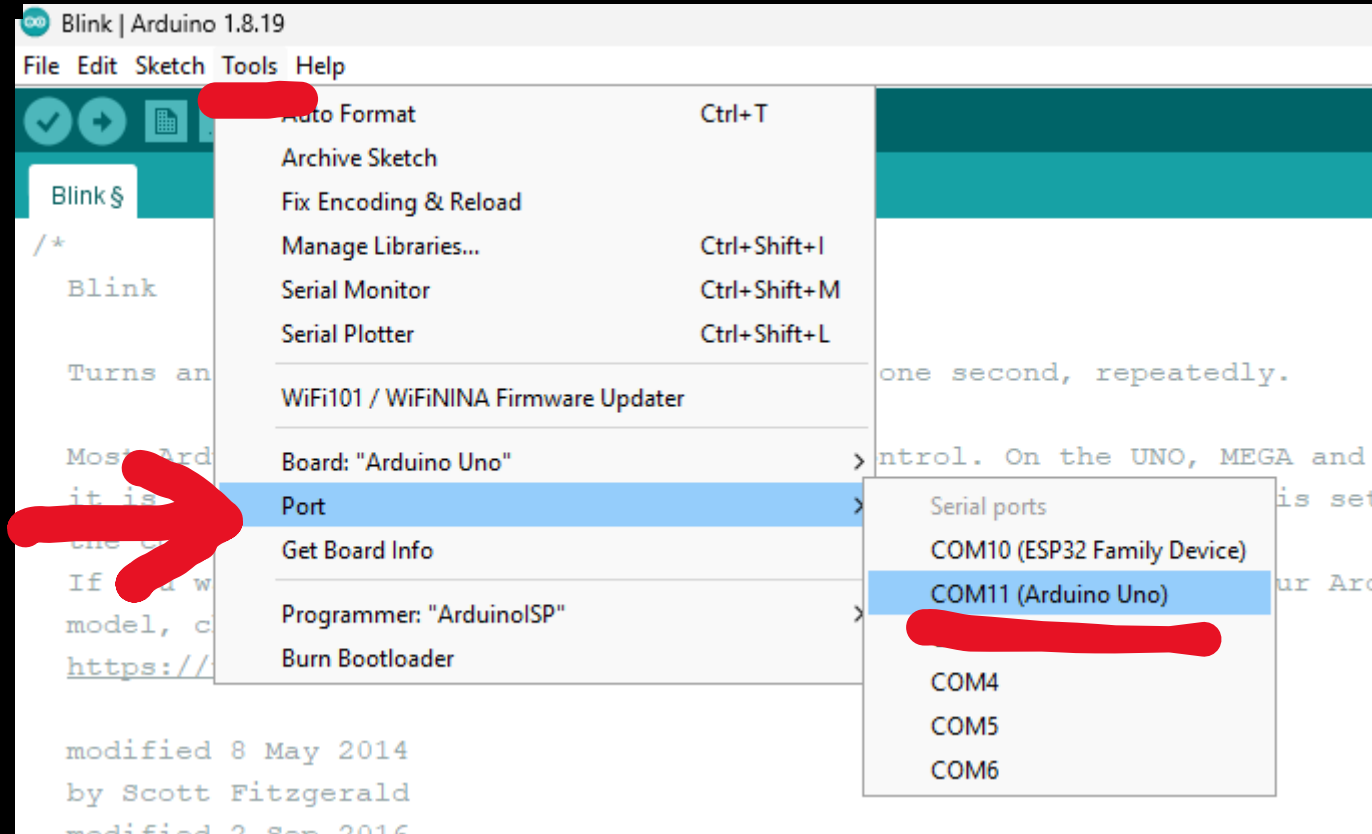
// the setup function runs once when you press reset or power the board
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(LED_BUILTIN, OUTPUT);
}

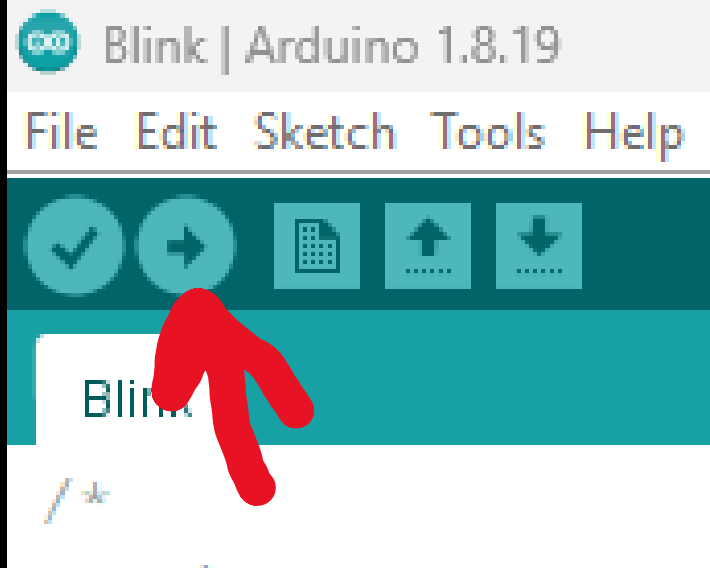
// the loop function runs over and over again forever
void loop() {
  digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)
  delay(1000); // wait for a second
  digitalWrite(LED_BUILTIN, LOW); // turn the LED off by making the voltage LOW
  delay(1000); // wait for a second
}
```

Plug in your Arduino <3

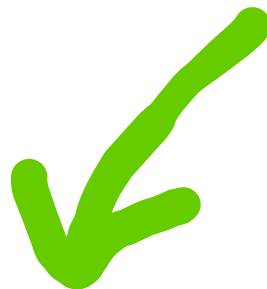








```
by using Arduino  
This example code is in the public domain.  
  
https://www.arduino.cc/en/Tutorial/BuiltInExamples/Blink  
*/  
  
// the setup function runs once when you press reset or power the board  
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}  
  
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void loop() {  
  digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)  
  delay(1000); // wait for a second  
  digitalWrite(LED_BUILTIN, LOW); // turn the LED off by making the voltage LOW  
  delay(1000); // wait for a second  
}  
  
Compiling sketch...  
  
C:\Program Files (x86)\Arduino\arduino-builder -dump-prefs -logger=machine -hardware C:\Pro  
  
28
```



Output

```
avrdude: AVR device initialized and ready to accept instructions

Reading | ##### | 100% 0.01s

avrdude: Device signature = 0x1e9801 (probably m2560)
avrdude: reading input file "C:\Users\Yung\AppData\Local\Temp\arduino\sketch
avrdude: writing flash (1536 bytes):

Writing | ##### | 100% 0.24s

avrdude: 1536 bytes of flash written

avrdude done.  Thank you.
```



BLINK



Demonstrate how to plug in battery!
Make sure the switch on the board is OFF
Double check the polarity.

<https://raw.flemingsociety.com/robot.txt>

COPY **ALL** code (CTRL + A > CTRL + C)

Replace all code on Arduino (CTRL + A > CTRL + V)

```
File Arduino 1.8.19
File Edit Shell Tools Help
[Icons]
[Menu]
// *****
// * This is the testing block for the robot
// * It contains several functions that can make the robot move around and sensor to sense value
// * The function includes:
// * - moving_forward()
// * - moving_backward()
// * - moving_left()
// * - moving_right()
// * - button_state()
// * - distance_measurement()
// * Try to play around with the blocks by putting them in the loop, and see what will happen
// * For the sensor value, you can read them in the serial monitor.
// * You can open it by pressing "ctrl + shift + M" to see the sensor value
// * Good luck and have fun playing around the robot!
// *
// *                               Junshu Chen, Jun. 21st 2023,
// *                               Updated by Yung Cheung Zhou, Jun. 25th 2025 *
// *****
// Constants (Do NOT change anything between the lines)
// *****
const int LEFT_BUTTON_PIN = 4;
const int RIGHT_BUTTON_PIN = 5;
const int FORWARD_MOTOR_PIN = 12;
const int FORWARD_MOTOR_SPEED = 3;
const int HORIZONTAL_MOTOR_PIN = 13;
const int HORIZONTAL_MOTOR_SPEED = 11;
const int ULTRASONIC_PIN = 6;
const int ULTRASONIC_TRIGGER_PIN = 7;
// Values that will be used in the program
int LEFT_BUTTON_STATE = 0;
int RIGHT_BUTTON_STATE = 0;
int DIRECTION_STATE = 0; // 0 means going to the right, 1 means going to the left
float duration_us, distance_cm;
unsigned long startMillis;
// *****
// Variables that will be used in the program
// Play around with the values below
// *****
const int OFFSET_TIME = 750; // Offset time in millisecond
const int FORWARD_SPEED = 50; // Set the value between 0 and 255
const int BACKWARD_SPEED = 20; // Set the value between 0 and 255
const int HORIZONTAL_SPEED = 50; // Set the value between 0 and 255
const int DISTANCE_LOWER_BOUNDARY = 4;
const int DISTANCE_UPPER_BOUNDARY = 5.5;
// *****
// Function prototypes (Do NOT change anything between the lines)
```

Find this

```
// The main loop of the program (Write your code here)
// *****
void loop() {
    // Put the function blocks here to make the robot move / read sensor values

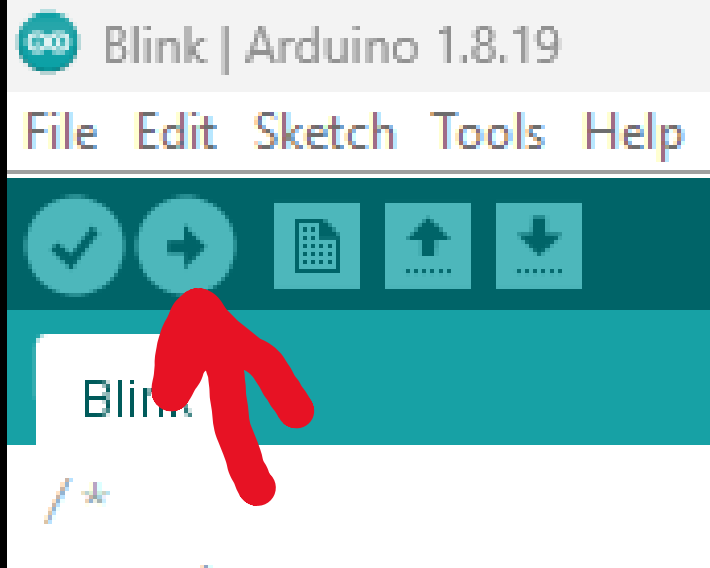
}
// *****
```

```
// The main loop of the program
// *****
void loop() {
    // Put the function blocks here
    distance_measurement();

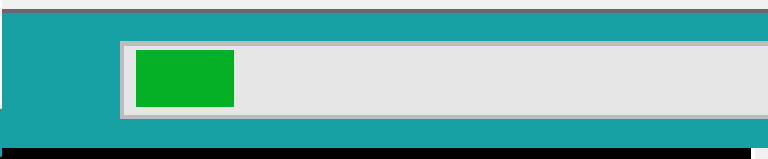
}
// *****
```

Type this:

distance_measurement();



```
by using Arduino  
This example code is in the public domain.  
  
https://www.arduino.cc/en/Tutorial/BuiltInExamples/Blink  
*/  
  
// the setup function runs once when you press reset or power the board  
void setup() {  
  // initialize digital pin LED_BUILTIN as an output.  
  pinMode(LED_BUILTIN, OUTPUT);  
}  
  
// the loop function runs over and over again forever  
void loop() {  
  digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)  
  delay(1000); // wait for a second  
  digitalWrite(LED_BUILTIN, LOW); // turn the LED off by making the voltage LOW  
  delay(1000); // wait for a second  
}
```

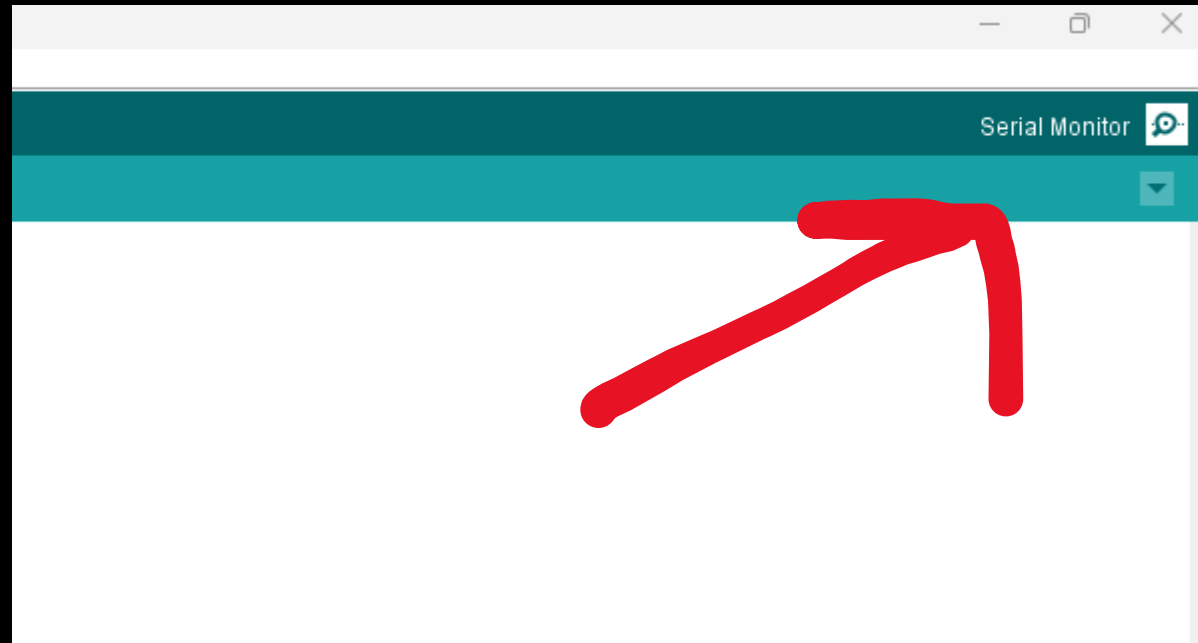


A screenshot of the Arduino IDE interface. The title bar reads "Blink | Arduino 1.8.19". The menu bar includes "File", "Edit", "Sketch", "Tools", and "Help". Below the menu bar is a toolbar with icons for a checkmark, a right arrow, a document, an upload arrow, and a download arrow. A red arrow points to the right arrow icon. Below the toolbar, the sketch name "Blink" is visible, followed by a code editor area showing the start of a C++ program with a comment "/*".

Compiling sketch...

C:\Program Files (x86)\Arduino\arduino-builder -dump-prefs -logger=machine -hardware C:\Pro

28



Now try :

`moving_forward();`

`moving_backward();`

`moving_left();`

`moving_right();`

`button_state();`

`distance_measurement();`

```
// The main loop of the program (Write your code here)
// *****
void loop() {
    // Put the function blocks here to make the robot move / read sensor values
    distance_measurement(); // MEASURES DISTANCE
    button_state(); // CHECKS THE BUTTON AND CHANGES DIRECTION IF NEEDED
    horizontal_movement(); // GOES HORIZONTALLY IN THE DIRECTION ABOVE
    forward_backward_movement(); // CHECKS IF NEED TO MOVE FORWARDS OR BACKWARDS
}
// *****
```

FINAL CODE

```
// Variables that will be used in the program
// Play around with the values below
// *****

const int OFFSET_TIME = 750;      // Offset time in millisecond
const int FORWARD_SPEED = 50;     // Set the value between 0 and 255
const int BACKWARD_SPEED = 20;    // Set the value between 0 and 255
const int HORIZONTAL_SPEED = 50;  // Set the value between 0 and 255
const int DISTANCE_LOWER_BOUNDARY = 4;
const int DISTANCE_UPPER_BOUNDARY = 5.5;
// *****
```

STUCK ON A CORNER?
YOU'RE THE ENGINEER!
CHANGE THESE!

CHALLENGES

MOVE DIAGONALLY TO SOLVE THE MAZE FASTER
PRE-EMPTIVE DIRECTION DECISION MAKING
(change directions after going through a gap)

LEADERBOARD

WINNER WINS A PRICE

Broke your robot?

Full code: <https://raw.flemingsociety.com/workingrobot.txt>